

GET ASPIRANT 2027

SOLIDWORKS CAD & MECHANICAL SYSTEMS

Precision 3D Modeling, Kinematic Mechanisms, Rapid Prototyping & Industrial Plant Optimization.

CAD ENGINE

SOLIDWORKS / Fusion 360

PROCESS IMPACT

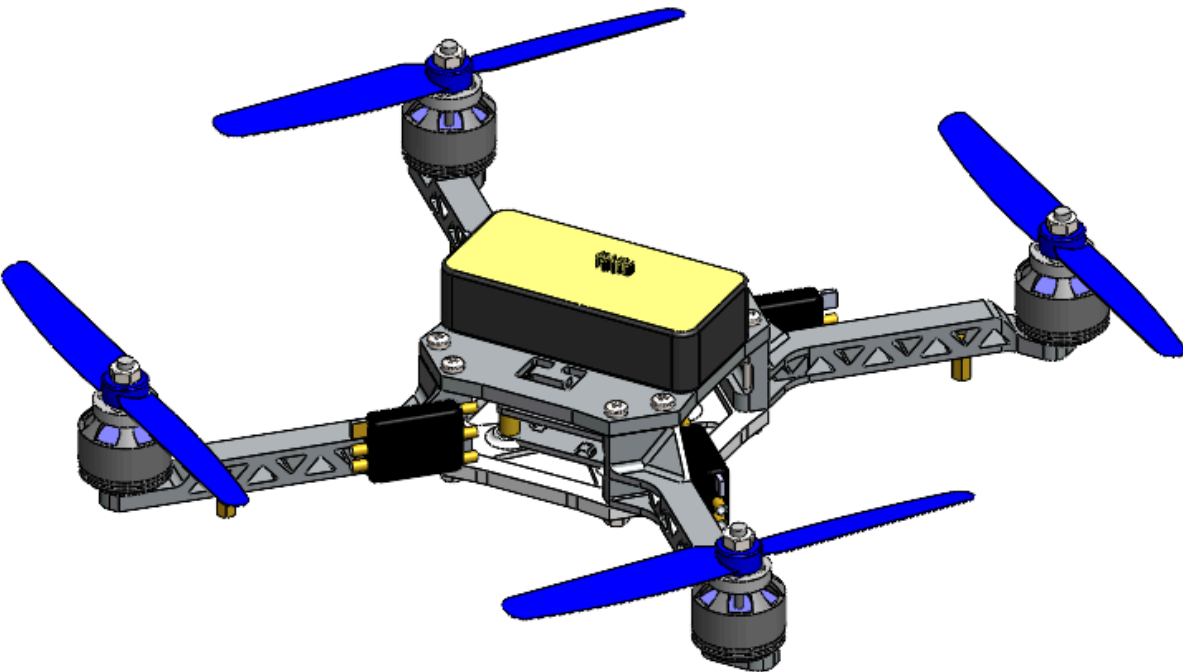
12-15% Cycle Time Reduction

FIELD EXPERIENCE

IOCL Loni LPG Bottling Plant

ASSEMBLIES

18+ Mated Components



HYGIENIC CIRCULAR APERTURE LID

The Problem: Traditional touch bins and pedal bins spread germs in public areas, and standard lid hinges break easily under constant use.

My Solution: Modeled an automatic circular aperture lid in SOLIDWORKS. The 8-blade mechanism opens smoothly in a circle, distributing movement evenly to extend motor life.

3D Printing Realization: Maintained a 0.3mm clearance gap between stacked leaves so parts print smoothly on FDM printers without binding.

CAD ENGINE

SOLIDWORKS Assembly

CLEARANCE

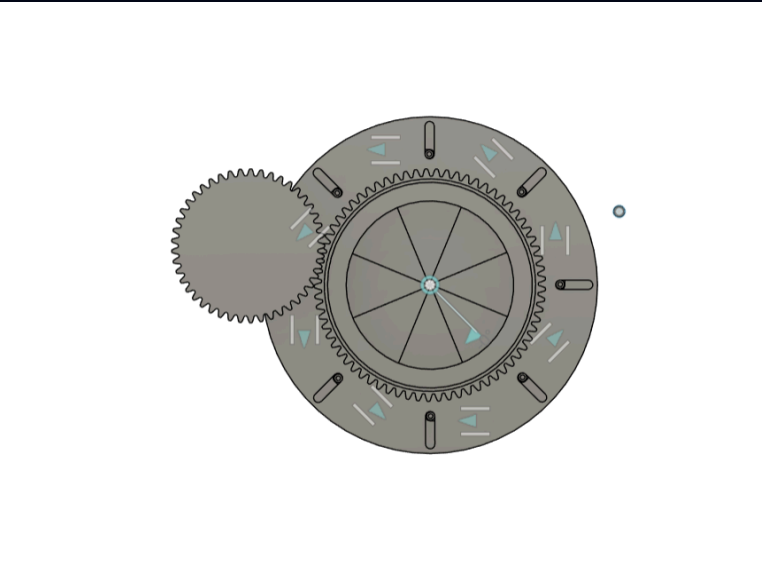
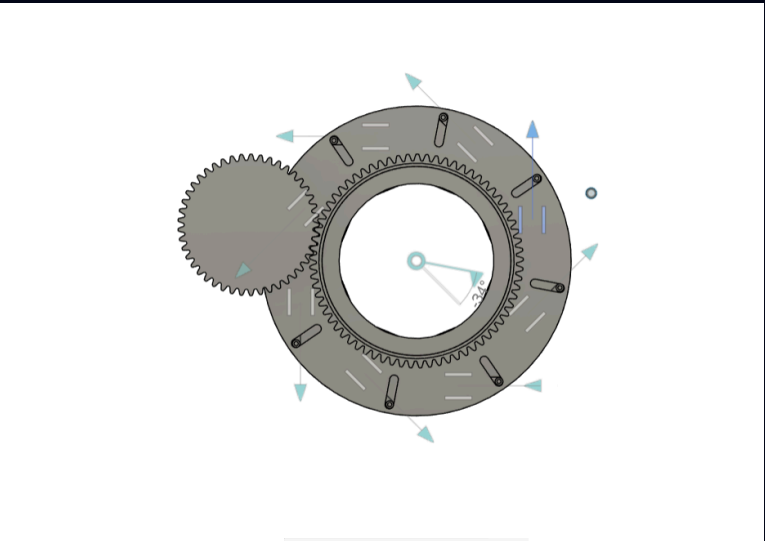
0.3 mm Tolerance

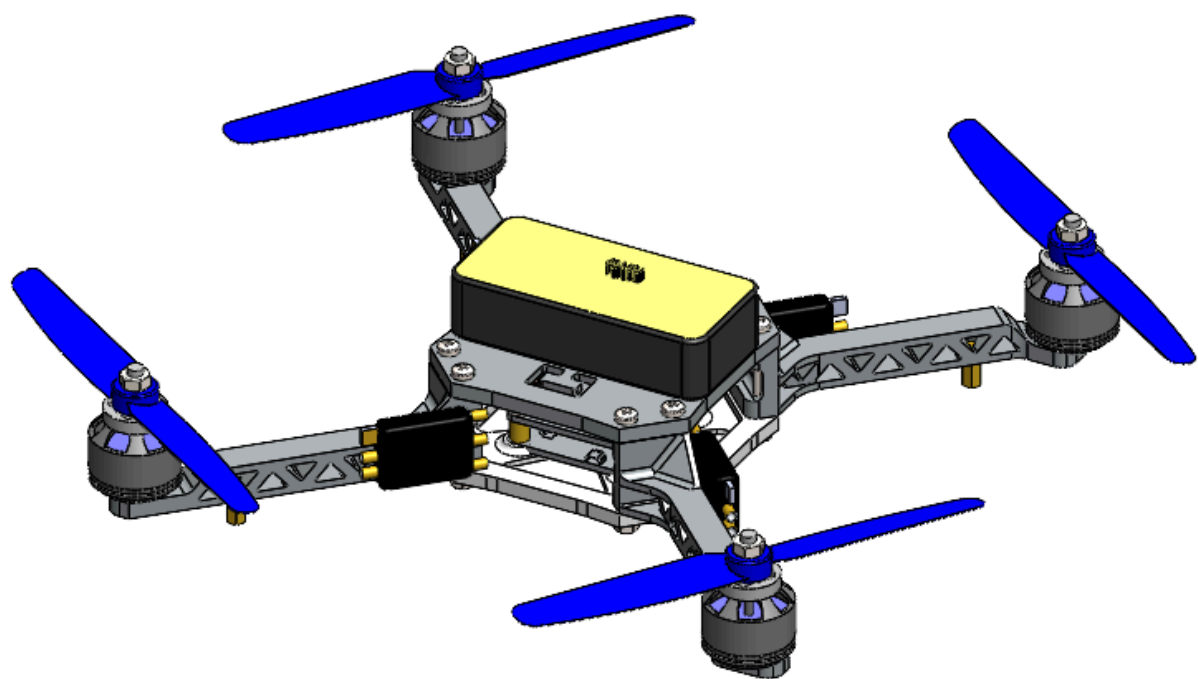
BLADES

8-Blade Synchronized

ACTUATOR

Outer Ring Pin Linkage





RIGID STRUCTURAL UAV FRAME

Objective: Modeled and assembled a complete F450 quadcopter drone frame in SOLIDWORKS by following professional video guides to master multi-part mates.

Engineering Execution: Modeled each component individually—central chassis plates, structural arm trusses, motor seats, brushless DC motors, and propeller blades—and assembled them with precise mating constraints (Concentric, Coincident).

Why It Matters: Teaches strict control over clearance management, screw alignment, and structural rigidity for production.

CAD SOFTWARE	COMPONENT COUNT
SOLIDWORKS Assembly	18+ Mated Parts
ARM DESIGN	ASSEMBLY MATES
Truss Rib Support	Concentric / Coincident

URBAN COMMUTER VEHICLE DESIGN

Project Context: Designed an urban electric scooter under TCG Design Studio during an automobile CAD internship.

Key Features: Engineered the tubular steel frame to house the lithium battery pack under the floorboard, keeping the weight low (Low Center of Gravity) for maximum riding stability.

Suspension & Rear Fit: Configured the rear swingarm and dual coil-spring shock absorbers with proper mudguard travel clearances.

CHASSIS TYPE

Tubular Steel Spaceframe

BATTERY HOUSING

Floorboard Low-CoG

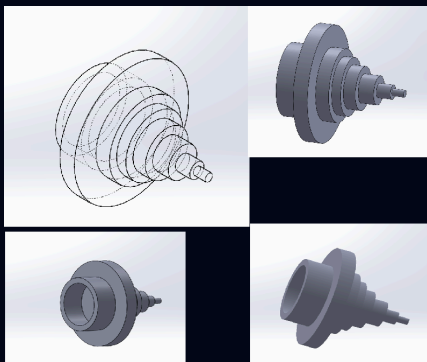
DRIVE UNIT

Rear Hub Motor Swingarm

SUSPENSION

Dual Coil Shock Clearance

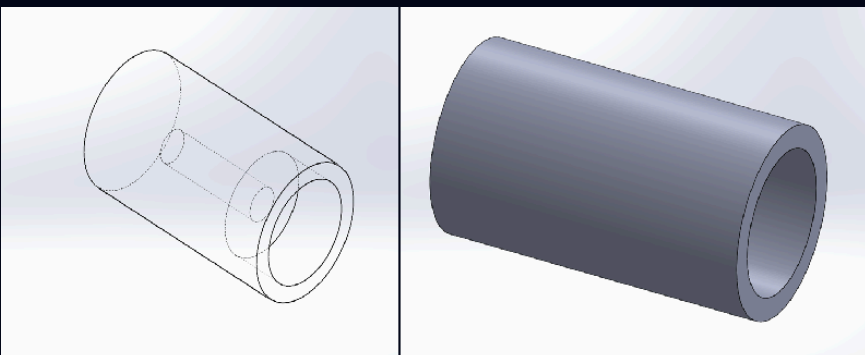




Multi-Step Tapered Sleeve

Modeled a multi-step external shaft collar with a hollow internal bore (180mm length, 7 taper steps, 40mm bore).

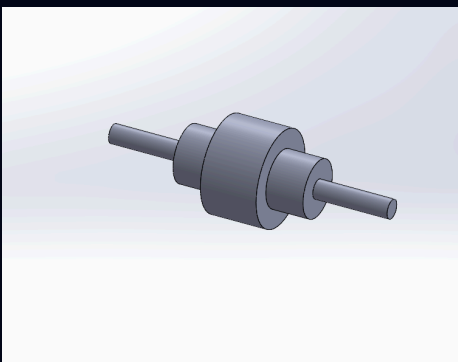
EXERCISE 1-3 // SOLIDWORKS



Hollow Stepped Cylinder

Modeled a hollow collar bushing featuring a two-stage internal stepped bore, mastering SECTION A-A interpretation.

EXERCISE 1-2 // SOLIDWORKS



Concentric Stepped Shaft

Modeled concentric cylindrical shoulders from standard engineering drawings, verifying 300mm axial shoulder spans.

EXERCISE 1-1 // SOLIDWORKS

INDIAN OIL CORPORATION LIMITED – INDANE LPG BOTTLING PLANT, LONI

⚡ 12% - 15% CYCLE TIME REDUCTION

Statutory Testing & Painting (STP) Facility Process Optimization

1. Pneumatic Clamping Fix

Reconfigured pneumatic line fittings and corrected Non-Return Valve (NRV) placements to eliminate pressure drops and clamping breakdowns.

2. Spray Booth Tuning

Adjusted spray booth pressure and nozzle angles to eliminate paint wastage and maintain BIS-mandated 60-micron total coating (30µm primer + 30µm enamel).

3. FIFO Material Layout

Proposed a trolley/conveyor transfer layout to eliminate batching delays between the Cold Repair Shed and inspection shed.

HYDRO TESTING

25 kg/cm² Water Pressure

QUALITY STANDARDS

PESO & BIS Guidelines

INSPECTION TOOLS

Ultrasonic Thickness Gauge

COATING THICKNESS

60 Microns (IS 2074 / 2932)

READY FOR GRADUATE ENGINEER TRAINEE (GET) ROLES

Combining Theoretical Mechanics, 3D CAD Precision & Plant Floor Experience

Seeking full-time Mechanical Engineering & Design opportunities upon 2027 graduation. Feel free to connect for interview discussions or project reviews.

EMAIL

rahmat888666@gmail.com

LOCATION

New Delhi, India

CAMPUS

DSEU GB Pant Okhla 1

LINKEDIN

md-rahmat-78a6a6220